E-WASTE CASE STUDIES

Case Studies 1 to 10

CASE STUDY 1

Why Japan Recycles More Than Most Countries

E-Waste Case Study Report

CASESTUDYREPORT

Why Japan Recycles More Than Most Countries

Lessons for India

Report prepared from publicly available sources; figures are as reported by the cited sources (checked September 2026).

1. Executive Summary

Japan is often held up as a recycling leader, and the reasons are institutional rather than cultural alone. Since the late 1990s it has layered product-specific laws on top of a national framework: a Containers and Packaging Recycling Law, a Home Appliance Recycling Law (in force since April 2001), and a 2012 law for small electronic devices. These laws make manufacturers responsible for recycling, make consumers pay at the point of disposal, and give municipalities a clear sorting and collection role.

The results are strong on collection and compliance: manufacturers have consistently beaten the legal recycling rates for the four regulated appliance types, and illegal dumping of appliances stayed at roughly 1.51.7% of units in the early years. However, the record is more mixed on plastics: about 87% of Japan's 8.24 million tonnes of plastic waste (2021) counts as "effectively used", but 63% of that is burned for energy recovery rather than recycled. India can borrow Japan's collection design, traceability and clear legal duties, while avoiding its heavy dependence on incineration.

2. Background

Japan is a densely populated, resource-poor island nation with limited landfill space. By the 1990s, rising waste volumes and scarce disposal sites forced policy action. The response was a "3R" (Reduce, Reuse, Recycle) approach built into national law, with separate statutes for different waste streams instead of a single generic rule.

The key instruments are: the Containers and Packaging Recycling Law (fully revised and enforced in April 2008, under which businesses pay money to municipalities that sort and collect packaging); the Specified Home Appliance Recycling Law (enacted 1998, enforced April 2001) covering air conditioners, televisions, refrigerators/freezers and washing machines/dryers; the Construction Materials Recycling Law (2000); and the Act on Promotion of Recycling of Used Small Electronic Devices (2012). The scheme covers different actors: consumers, retailers, manufacturers and local governments each have a defined duty. **3. Strategy / Initiative**

Japan's strategy rests on clear role allocation, physical infrastructure close to citizens, and traceability:

- Product-specific laws: separate rules for packaging, home appliances, construction materials and small electronics, each tailored to the material stream.
- Manufacturer duty with legal recycling targets: producers must take back and recycle regulated appliances and meet minimum weight-based recycling rates (for example 60% for air conditioners and

50% for refrigerators and washing machines when the law started), plus recover and destroy or recycle CFC/HCFC/HFC refrigerants.

- Consumer pays at disposal, retailers collect: households pay a fee when discarding a regulated appliance and hand it to the retailer or to a designated collection site (about 379 sites), which reduces the incentive to dump.
- Traceability: a "recycling tag" system lets consumers follow their discarded appliance online, and manufacturers publish annual recycling results.
- Municipal sorted collection: local governments run sorted collection of glass, cans, PET bottles, paper and plastics, supported financially by industry under the packaging law.
- Policy correction on plastics: a 2019 policy package targets plastic waste and greenhouse-gas emissions from burning waste, and Tokyo has announced a 40% cut in plastic burning by 2030.

4. Impact & Results

Table 1 summarises the main measured outcomes. Figures come from Japanese government and industry sources for the years shown; the periods differ because different reporting series are used.

Indicator	Result	Year / Source
Regulated appliances collected at designated sites	8.55 million units rising to 11.22 million units	FY2001 to FY2004, METI / AEHA
Illegally dumped appliances (share of units)	About 1.5-1.7% across the four years	FY2001-FY2004, METI
Air conditioners: recycling rate vs legal minimum	84% vs 60% required	FY2005, AEHA
Televisions: recycling rate vs legal minimum	77% vs 55% required	FY2005, AEHA
Refrigerators: recycling rate vs legal minimum	66% vs 50% required	FY2005, AEHA
Washing machines: recycling rate vs legal minimum	75% vs 50% required	FY2005, AEHA
PET bottle recycling rate	72.4%	FY2010, Japan Govt
Steel cans / aluminium cans / cardboard	89.4% / 92.6% / 99.3%	FY2010, Japan Govt
Plastic waste: total and share effectively used	8.24 Mt; 87% used (25% recycled, 63% energy recovery)	2021, MOEJ
Plastic waste not used	13% (8% burned without energy recovery, 5% landfilled)	2021, MOEJ

Interpretation: Japan's numbers show that a well-designed producer-responsibility system can push actual performance well above the legal floor. The weakness is visible in the plastics row: a high "effective use" figure hides the fact that most of it is thermal recovery. Critics note that burning plastic emits greenhouse gases and that some waste is exported, so the headline recycling rate should be read with care.

5. SWOT Analysis

Strengths	Weaknesses
• Clear legal duties for every actor • High collection and compliance rates • Recycling-tag traceability • Dense collection network and strong household sorting habit	• Heavy reliance on thermal recovery for plastics • Recycling-rate definitions differ from EU WEEE, so comparisons are tricky • Sorting is time-consuming for citizens
Opportunities	**Threats**
• Shift to chemical and material recycling • Design for recycling and reuse • Urban mining of metals from small electronics	• Climate concerns about incineration • Export of waste to countries with weak infrastructure • Falling waste volumes may leave recycling plants under-used

6. Comparison with India

The gap between Japan and India is less about rules on paper and more about collection, incentives and enforcement.

Dimension	Japan	India
Legal design	Separate laws per stream (appliances, packaging, small electronics)	E-Waste (Management) Rules 2022, SWM Rules 2016, Plastic Waste Rules; broad but enforcement is patchy
Who pays	Consumer pays a fee at disposal; manufacturer must recycle to legal rates	Producers buy EPR certificates; consumers usually get paid by scrap dealers, which feeds the informal sector
Collection	About 379 designated sites plus retailer take-back	Over 90% of e-waste reportedly handled informally through kabadiwalas
Traceability	Recycling tag and published manufacturer results	Online CPCB EPR portal introduced from 2023; still maturing
Household behaviour	Strong sorting habit backed by municipal schedules	Segregation is mandatory but uneven (about 43% segregated even in Pune)
Weak point	Plastic incineration and exports	Informal recycling with unsafe methods, low formal capacity utilisation

7. Recommendations

1. Introduce a visible end-of-life fee or deposit at purchase or disposal so that consumers, not just producers, take part in the system.
2. Make retailers and service centres statutory take-back points for regulated appliances, and publish a locator for citizens.

3. Adopt a Japanese-style recycling tag or QR trace so each collected appliance can be tracked to a registered recycler.
4. Set recycling targets by product category (not only by metal recovery) and audit them, as Japan does for its four appliance types.
5. Pilot chemical and mechanical plastic recycling before scaling waste-to-energy, to avoid locking India into Japan's incineration dependence.
6. Integrate the informal sector as collection partners with training, rather than trying to replace it.

8. Conclusion

Japan's success comes from a system in which each actor has a defined role, collection is convenient, and results are measured and published. For India, the transferable lessons are take-back convenience, traceability and category-wise targets. The cautionary lesson is that a high headline recycling number can conceal heavy thermal recovery. India should design its systems around real material recovery from the start, and pair strong regulation with the informal workforce that already exists.

9. References

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ONE-PAGE EXAM REVISION SHEET | Case 1: Why Japan Recycles More Than Most Countries

Must-remember facts & numbers

- Home Appliance Recycling Law: enacted June 1998, enforced April 2001; covers AC, TV, fridge/freezer, washing machine/dryer.
- Small electronics law: 2012. Containers & Packaging law fully revised April 2008.
- Legal minimum rates at start: AC 60%, TV 55%, fridge 50%, washing machine 50%; FY2005 actual 84 / 77 / 66 / 75%.
- Collected appliances: 8.55 M (FY2001) to 11.22 M (FY2004); illegal dumping about 1.5-1.7%.
- Plastics 2021: 8.24 Mt; 87% "effective use" = 25% recycled + 63% energy recovery; 13% unused.
- Tokyo target: cut plastic burning 40% by 2030.

Key terms

3R	Reduce, Reuse, Recycle; basis of Japan's recycling policy.
Thermal recovery	Burning waste (often plastic) to recover energy; counted as "used" in Japan but not true recycling.
Recycling tag	System letting consumers trace their discarded appliance online.
Designated collection site	Manufacturer-run drop-off sites (about 379) for regulated appliances.
EPR	Extended Producer Responsibility: producer is responsible for the product at end of life.

Likely exam questions (frame your answer around these)

1. Explain the key features of Japan's appliance recycling system and why compliance is high.
2. Critically evaluate Japan's "high recycling rate" for plastics.
3. What can India learn from Japan? Suggest three concrete measures.

Answer skeleton

Laws by stream -> clear roles (consumer pays, retailer collects, manufacturer recycles) -> traceability -> results with numbers -> limitation (thermal recovery) -> India lessons.

Memory hook: *"Pay, Drop, Trace, Recycle": consumer pays, drop at retailer/site, trace by tag, manufacturer recycles.*

CASE STUDY 2

Switzerland's E-Waste Success Story

E-Waste Case Study Report

CASE STUDY REPORT

Switzerland's E-Waste Success Story

A private, fee-funded take-back system with national reach

Report prepared from publicly available sources; figures are as reported by the cited sources (checked September 2026).

1. Executive Summary

Switzerland was among the first countries to organise electronic waste take-back. Its system, run mainly by two industry bodies (SENS eRecycling for household appliances and lighting, and Swico Recycling for IT, telecom and consumer electronics), is financed by an advance recycling fee added to the price of new products. Because the fee is paid up front, citizens can return old devices free of charge to shops or collection points.

The outcome is one of the highest per-capita collection levels in the world: SENS alone reported about 11.14 kg of equipment per person in 2024, and about 1.4 million tonnes have been collected since 1990. The system works because of convenience, an up-front financing model, accredited recyclers within Switzerland and strict control over exports. For India, the key lesson is that funding, convenience and enforcement must work together; the current Indian design does not yet have an equivalent up-front fee.

2. Background

Switzerland is a small, wealthy, high-consumption country with limited landfill space and strong environmental regulation. Industry initiated e-waste take-back early: SENS was founded in 1990 and later became the largest of the take-back systems. A federal ordinance on the return, take-back and disposal of electrical and electronic equipment (in force from 1998) gave the system a legal base: consumers must return old appliances and retailers, manufacturers and importers must take them back.

Two organisations run the system in parallel. SENS handles household appliances (refrigeration, large and small household appliances, lighting, photovoltaic modules), while Swico handles IT, office, telecom and consumer electronics. In 2016 the two together handled about 137,808 tonnes, of which 62% was household appliances and lamps (SENS) and 38% was IT and consumer electronics (Swico).

3. Strategy / Initiative

The Swiss model can be summarised as "pay in advance, return anywhere, recycle locally":

- Advance recycling fee (ARF): producers and importers pay a per-unit fee in Swiss francs on every product sold; SENS manages the fund, which pays for collection, transport, treatment and system management.
- Free return: consumers can return equipment to any retailer that sells that type of product or to a collection point, without having to buy a new device.
- Dense network: over 750 accredited collection points (including municipalities) were reported for 2021, with about 630 SENS points listed in current material.

- Accredited recyclers: 16 recycling companies process SENS equipment, all in Switzerland (only photovoltaic modules go to nearby countries).
- Voluntary participation, backed by law: about 1,250 producer, importer and retailer partners participate in SENS contracts, with government limiting inspections as long as the private sector fulfils its duties.
- Export control: exporting defective appliances requires a permit from the Federal Office for the Environment, which limits illegal shipments.

4. Impact & Results

Table 1 shows selected performance data from SENS and Swico. SENS figures include lighting, photovoltaics and (from some years) batteries and other items, so year-to-year comparison should be made carefully.

Indicator	Result	Year / Source
Total collected, SENS + Swico	137,808 tonnes	2016, Swiss environment reporting
Split between the two systems	62% household appliances and lamps; 38% IT and consumer electronics	2016
SENS quantity processed	About 95,800 tonnes	2021, SENS factsheet
SENS total collected	89,960 tonnes (10.23 kg per person)	2022, SENS
SENS total collected	92,874 tonnes (10.36 kg per person)	2023, SENS
SENS collection per capita	11.14 kg	2024, SENS
Cumulative collection	More than 1.4 million tonnes since 1990	SENS website
Average material recovery from an item	About 75% reused as material	SENS
Producer / importer / retailer partners	About 1,250	1 Jan 2022, SENS

Interpretation: high collection is achieved without deposit-refund complexity, mainly because return is free and easy, and the fee has already been paid. SENS points out that its collection figures do not count devices stored at home or passed on for reuse, so the true recovery rate is likely higher than measured. The system also depends on the reliability of the fee: the ARF works like a pay-as-you-go pension fund, meaning that today's sales finance tomorrow's recycling.

5. SWOT Analysis

Strengths	Weaknesses
• Up-front ARF guarantees funding • Free and convenient return via shops and 600+ points • Recycling done inside Switzerland with audited recyclers • Long track record since 1990	• Item-to-kilogram conversions cause measurement inaccuracies • Unused stored devices are not counted • Voluntary base relies on continued industry cooperation
Opportunities	**Threats**
• Growing streams: solar panels, heat pumps, ecigarettes • Better repair and reuse channels • Improved material recovery for critical metals	• Small devices and cables thrown in general waste • Online sellers outside the fee system • Rapid growth of new device types with different recycling needs

6. Comparison with India

Switzerland shows what a mature, fee-funded system looks like. India's system is younger, larger, and dominated by informal collection.

Dimension	Switzerland	India
Financing	Advance recycling fee paid on every sale, held by SENS/Swico	EPR certificates bought from recyclers; price band linked to environmental compensation (floor about Rs 22/kg)
Return convenience	Free return to any seller or 600+ collection points	Limited formal take-back points; consumers often sell to scrap dealers for cash
Scale of generation	About 10-11 kg per person collected through SENS	About 1.78 million tonnes generated in 2023-24, roughly 1.2-1.3 kg per person (author calculation)
Who recycles	16 accredited recyclers, all in Switzerland	322 registered recyclers (Feb 2025) but most e-waste still flows to the informal sector
Export / import control	Permit required for exporting defective appliances	Illegal imports and informal trade routes remain a concern
Enforcement style	Industry-run, with light state oversight	State-run portal with audits; capacity constrained

7. Recommendations

1. Explore a visible advance recycling contribution or deposit on electronics so recycling is pre-funded and free at the point of return.
2. Require all retailers (including e-commerce platforms) to accept old devices of the same type free of charge.

3. Establish a small number of highly visible, accredited collection points in every district, starting with cities.
4. Publish annual collection and recovery data per producer group, including measured stock and reuse, as SENS does.
5. Bring informal aggregators into the accredited network so they receive a fee for collection rather than competing with it.

8. Conclusion

Switzerland's e-waste success is a systems success: funding is guaranteed by the advance fee, consumers face no cost or effort, recyclers are accredited, and exports are controlled. India cannot copy the model exactly because of its scale and informal economy, but it can adopt the core principle: make the right behaviour the easiest and cheapest one for the consumer.

9. References

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ONE-PAGE EXAM REVISION SHEET | Case 2: Switzerland's E-Waste Success Story

Must-remember facts & numbers

- SENS founded 1990; Swico handles IT/telecom/consumer electronics; SENS handles household appliances, lamps, PV.
- Financing: advance recycling fee (ARF) on every unit sold, managed by SENS; return is free.
- 2016: 137,808 t total (62% SENS-type, 38% Swico-type). SENS 2022: 89,960 t (10.23 kg/person); 2023: 92,874 t (10.36 kg); 2024: 11.14 kg.
- More than 1.4 Mt collected since 1990; ~75% material reuse; 16 recyclers, all in Switzerland.
- About 1,250 partners (2022); 750+ accredited collection points (2021 factsheet).

  Defective appliance exports need Federal Office for the Environment permit.

Background (1990, ordinance) -> ARF financing -> free return + dense network -> local accredited recyclers -> data -> India comparison -> transferable vs non-transferable.

Memory hook: *"Fee first, return free, recycle at home."*

Key terms

ARF	Advance recycling fee: paid at sale, funds later collection and recycling.
SENS / Swico	Swiss industry take-back organisations (appliances / IT & telecom).
Collection rate	Collected e-waste relative to new equipment put on the market.
Take-back obligation	Retailers must accept old devices free of charge.
Pay-as-you-go model	Today's fees pay for today's recycling, like a pension scheme.

Likely exam questions (frame your answer around these)

1. Describe how the Swiss e-waste system is financed and why it works.
2. Compare the Swiss ARF with India's EPR certificate system.
3. Which elements of the Swiss model are transferable to India, and which are not?

Answer skeleton

CASE STUDY 3

India's E-Waste (Management) Rules, 2022

E-Waste Case Study Report

CASESTUDYREPORT

India's E-Waste (Management) Rules, 2022

From voluntary compliance to a certificate-based EPR regime

Report prepared from publicly available sources; figures are as reported by the cited sources (checked September 2026).

1. Executive Summary

The E-Waste (Management) Rules, 2022 were published on 2 November 2022 and came into force on 1 April 2023. They replaced the 2016 Rules and shift India from broad obligations to a measurable, portal-based Extended Producer Responsibility (EPR) system. Producers must meet annual recycling targets by buying EPR certificates from registered recyclers, with all entities registering on a CPCB online portal.

The Rules cover 106 items of electrical and electronic equipment (up from 21), add recycling targets rising from 60% to 80%, introduce refurbishing certificates and a restriction on hazardous substances (RoHS), and provide for environmental compensation. The design is ambitious, but early results show heavy dependence on enforcement quality: e-waste generation grew 151% between 2017-18 and 2023-24, and most is still believed to be handled informally.

2. Background

India is among the largest e-waste generators in the world. CPCB data show generation rising from about 7.08 lakh tonnes in 2017-18 to about 17.78 lakh tonnes in 2023-24. Earlier estimates were 13.46 lakh tonnes in 2020-21 and 16.01 lakh tonnes in 2021-22. India's first e-waste rules appeared in 2011 and were revised in 2016; EPR was introduced in 2012 but compliance was largely voluntary and, according to industry participants, not taken seriously.

Weak enforcement, reliance on informal dismantling and the lack of a verifiable link between producer obligations and actual recycling led the government to redesign the framework in 2022. Internationally, only a small share of global e-waste is formally collected and recycled (17.4% in 2019 according to the Global Ewaste Monitor), which is the wider context for India's challenge.

3. Strategy / Initiative

The Rules rest on six main mechanisms:

- Wider coverage: 106 categories of EEE from 2023-24, including many household, IT, and telecom items, plus bulk consumers who must dispose through authorised recyclers or refurbishers.
- Mandatory registration: producers, manufacturers, recyclers and refurbishers must register on the CPCB EPR portal; unregistered entities cannot lawfully operate.
- Recycling targets by weight: 60% of estimated waste (products sold in year Y minus the average product life) in 2023-24 and 2024-25, 70% in 2025-26 and 2026-27, and 80% from 2027-28 onward.

- EPR certificates: certificates are generated for registered recyclers based on verified recovery of key end products (gold, copper, aluminium, iron), and producers buy them to meet targets; purchase is capped at current liability plus past shortfall plus 5%.
- Refurbishing certificates: these defer, rather than cancel, producer obligations for the extended life of a refurbished product, to promote reuse.
- Price control and penalties: the 2024 amendment set the certificate price between 30% and 100% of environmental compensation (a floor of about Rs 22/kg for most items), and non-compliance attracts environmental compensation and prosecution under the Environment (Protection) Act 1986. RoHS limits apply for lead, mercury, hexavalent chromium, certain brominated flame retardants (0.1%) and cadmium (0.01%).

4. Impact & Results

The Rules are recent, so impact data mostly show system build-up rather than final environmental outcomes.

Indicator	Result	Year / Source
E-waste generation	7,08,445 t to 17,78,400 t (+151%)	2017-18 to 2023-24
E-waste generation (earlier series)	13,46,496 t (2020-21); 16,01,155 t (2021-22)	Lok Sabha reply, CPCB
EEE categories covered	106 (previously 21)	From 2023-24
Registered producers	7,226	10 Dec 2024, CPCB report to NGT
Registered recyclers / refurbishers	295 / 53 (Dec 2024); 322 / 72 (Feb 2025)	CPCB; Rajya Sabha reply
Reported recycler capacity	About 22.09 lakh tonnes per year	9 Feb 2025, CPCB
Producers registered in 2023-24	4,868 (of which 2,209 with nil EPR target per CPCB)	CPCB report
Recycling target path	60% (2023-25), 70% (2025-27), 80% (2027-28 onward)	Schedule III
Certificate price floor	About Rs 22 per kg (band 30-100% of compensation)	2024 Amendment

Interpretation: registered capacity (about 22 lakh tonnes) is larger than estimated generation, yet an estimated 90% of e-waste is still handled by the informal sector, so the issue is collection and enforcement rather than plant capacity. The price floor is contested: several large appliance makers challenged it in the Delhi High Court from 2024-25, and reports describe interim relief in one case but no final judgment as of mid-2026.

5. SWOT Analysis

Strengths	Weaknesses
• Legally binding, target-based EPR • Central online portal and public dashboard • Coverage of 106 items plus bulk consumers • Refurbishing certificates encourage reuse	• Limited audit and tracking capacity at CPCB • Certificates measured only on four metals • Little focus on repair as a preventive step • Heavy compliance load on small producers
Opportunities	**Threats**
• Formalising the informal sector • Urban mining and critical-metal recovery • Growth of a recycling industry and green jobs • Digital traceability with QR/blockchain tools	• Fraudulent or paper certificates • Litigation over price floors • Continued illegal imports and informal recycling • Rapid growth of e-waste volumes

6. Comparison with India

This case is about India itself, so the useful comparison is between the design of the 2022 Rules and the benchmark systems in Japan and Switzerland studied in this course.

Dimension	Rules 2022 (design)	India
Funding logic	Producers buy certificates from recyclers (market-based)	Switzerland: fee at sale pre-funds recycling; Japan: consumer fee at disposal
Consumer role	Bulk consumers regulated; households not directly	Both benchmarks: consumers have clear duty to return
Target basis	Weight-based, using sales minus average product life	Japan: legal weight-based rates per appliance type
Traceability	CPCB EPR portal and certificate tracking	Japan: recycling tag; Switzerland: audited accredited recyclers
Enforcement capacity	Constrained; audits and penalties still developing	Strong routine oversight and high compliance culture
Informal sector	Not directly integrated in the rules	Not a major issue in either benchmark country

7. Recommendations

1. Strengthen CPCB auditing with third-party verification and data-matching (sales data, recycler input and output, certificate issue).
2. Give formal roles to informal collectors through registration, training and payment for delivering ewaste to authorised recyclers.
3. Add household-facing take-back requirements for retailers and online platforms.
4. Widen certificate measurement beyond four metals to include plastics, glass, and hazardous components handled safely.

5. Publish annual outcome data (tonnes collected, recycled and recovered per category), and settle the price-floor question transparently.

8. Conclusion

The 2022 Rules are a substantial upgrade: they define targets, create a market for certificates and require registration of all actors. Their success will depend on whether the system can verify that certificates represent real recycling and whether it can draw the large informal sector into the formal chain. The framework is credible on paper; the next few years will show whether enforcement can match the ambition.

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ONE-PAGE EXAM REVISION SHEET | Case 3: India's E-Waste (Management) Rules, 2022

Must-remember facts & numbers

- Published 2 Nov 2022; in force 1 Apr 2023; replaced 2016 Rules.
- Coverage: 106 EEE items (from 21); bulk consumers included.
- Targets: 60% (2023-25), 70% (2025-27), 80% (2027-28 on).
- EPR certificates from registered recyclers; key metals Au, Cu, Al, Fe; purchase cap = liability + past shortfall + 5%.
- Price floor about Rs 22/kg (2024 amendment; 30-100% of environmental compensation).
- Generation 7.08 lakh t (2017-18) to 17.78 lakh t (2023-24), +151%; recycler capacity about 22.09 lakh t/yr (Feb 2025).

Background (growth, weak 2016 rules) -> six mechanisms -> data table -> what works vs gaps (enforcement, informal) -> comparison -> recommendations.

Memory hook: *"Register, Target, Certify, Audit": the four steps of the 2022 regime.*

Key terms

EPR	Extended Producer Responsibility.
EPR certificate	kg-denominated proof of recycling generated by a registered recycler and bought by producers.
Refurbishing certificate	Defers (does not cancel) EPR for the extended life of a refurbished product.
Environmental compensation	Penalty for non-compliance; also sets the price band for certificates.
RoHS	Restriction of Hazardous Substances limits (0.1% lead etc.; 0.01% cadmium).

Likely exam questions (frame your answer around these)

1. Explain the major provisions of the E-Waste (Management) Rules 2022.
2. How do EPR certificates work? What are their limitations?
3. Evaluate the 2022 Rules against the Swiss or Japanese model.

Answer skeleton

CASE STUDY 4

India's Extended Producer Responsibility (EPR)

E-Waste Case Study Report

CASESTUDYREPORT

India's Extended Producer Responsibility (EPR)

Success or paper compliance?

Report prepared from publicly available sources; figures are as reported by the cited sources (checked September 2026).

1. Executive Summary

Extended Producer Responsibility makes producers financially and operationally responsible for products at end of life. India's e-waste EPR was largely voluntary in 2012-2016; from 2023 it became a certificate-based, portal-tracked system with legal targets. The question for this case is whether that system produces real recycling or mainly generates paperwork.

The evidence points to a mixed answer. On the positive side: thousands of producers registered, hundreds of recyclers with about 22 lakh tonnes of reported capacity, and a public certificate market. On the negative side: an estimated 90% of e-waste is still processed informally, a contested price floor (Rs 22/kg against market bids reported as low as Rs 5.90/kg), thin regulatory capacity and documented certificate fraud in the related plastic EPR system. The fair verdict is "real architecture, uneven verification", which means paper compliance is a genuine risk unless audit and traceability improve.

2. Background

EPR entered Indian e-waste policy in 2012 on a voluntary basis, which industry participants describe as ineffective because producers had little incentive to act. The 2016 Rules made EPR mandatory but left the mechanism vague. The 2022 Rules introduced a detailed EPR framework: producers register, receive targets and fulfil them by buying certificates generated by registered recyclers and refurbishers through the EPR portal.

The scale of the challenge is large. E-waste generation reached about 17.78 lakh tonnes in 2023-24, and India's formal collection remains small relative to that. In CPCB's December 2024 figures, 7,226 producers, 295 recyclers, 35 manufacturers and 53 refurbishers were registered, and of the 4,868 producers registered in 2023-24, 2,209 were reported with nil EPR targets.

3. Strategy / Initiative

The EPR strategy is built on market and compliance instruments:

- Obligation set by CPCB: each producer is assigned a target based on past sales and product life.
- Certificate trading: registered recyclers earn kilogram-denominated certificates for verified recovery; producers purchase to meet targets; electronic trading platform guidelines were issued in January 2025.
- Price band: the 2024 amendment set the certificate price between 30% and 100% of environmental compensation, giving a floor near Rs 22/kg.
- Verification and audit: registrations, returns and dashboards on the CPCB portal, with provisions for audits and penalties.

- Reuse pathway: refurbishing certificates defer producer obligations to reward repair and life extension.

4. Impact & Results

The table combines official and reported figures. Some are estimates from media and think-tank reports and should be read as indicative rather than audited.

Indicator	Evidence	Source / Year
Producers registered	7,226 (Dec 2024)	CPCB report to NGT
Recyclers registered	295 (Dec 2024); 322 (Feb 2025)	CPCB; Rajya Sabha
Reported recycler capacity	About 22.09 lakh t/yr	Rajya Sabha, Feb 2025
Generation	17.78 lakh t (2023-24)	CPCB via reports
Share handled informally	About 90%	Editorial summaries, 2025
Price floor vs market	Floor Rs 22/kg; some market bids reported at Rs 5.90/kg	Media analysis, 2025
CPCB budget (2025-26)	Rs 126 crore, viewed as insufficient	Editorial analysis
Compensation disbursed for environmental use	About 1% of amount collected (reported)	Editorial analysis
Fake certificates (plastic EPR)	Over 600,000 fake certificates found in a 2023 audit	Drishti IAS summary of CPCB audit
Legal challenge	OEMs challenged the price band in Delhi High Court; case ongoing	2025-2026 reports

Interpretation: registration counts and certificate volumes show activity, not outcomes. The gap between reported capacity (about 22 lakh tonnes) and estimated generation (about 17.8 lakh tonnes) would suggest that recycling capacity is not the bottleneck; feedstock collection and verification are. The fake-certificate finding relates to plastic EPR, not e-waste, but it illustrates how a certificate market can be gamed if inputs are not physically verified.

5. SWOT Analysis

Strengths

- Statutory targets and public portal
- Large registered recycler base
- Refurbishing option supports reuse
- Producers now have quantified liabilities

Weaknesses

- Weak audit, tracking and penalties
- Certificates measured on limited metals
- Price floor disputes and market distortion
- Informal sector excluded from the value chain

Opportunities	Threats
• Third-party audits and digital traceability • Design-for-recycling and repair incentives • Formalising kabadiwalas as collection agents • Public reporting to build trust	• Certificate fraud and paper recycling • Litigation and policy uncertainty • Informal recyclers undercutting formal ones • Rapid volume growth

6. Comparison with India

Comparing the Indian EPR system with the Swiss (fee-funded) and Japanese (consumer-pays) designs shows where the Indian model is most exposed.

Dimension	Global good practice	India
Funding	Producer-financed through market-priced certificates	Swiss ARF pre-funds recycling; Japan uses consumer fee at disposal
Verification	Portal-based, limited physical audit	Accredited recyclers and audited take-back (Switzerland); manifest and recycling tag (Japan)
Collection	Left largely to recyclers and market	Legally mandated retailer take-back and convenient drop-off
Consumer incentive	Weak; scrap dealers pay cash	Free or pre-paid return
Result signal	Registrations and certificate volumes	Published tonnes collected per capita and recycling rates

7. Recommendations

1. Link certificates to physical evidence: weighbridge data, GPS-tracked transport and third-party verification of recycler inputs and outputs.
2. Publish independent annual audits of certificate issuance versus collected quantities.
3. Replace or clarify the price floor with a transparent, evidence-based mechanism, after considering recycler costs and producer burden.
4. Pay formal collection incentives to informal aggregators and integrate them through registration.
5. Build regulator capacity: adequate CPCB and state board staffing and budget, and enforce environmental compensation.

8. Conclusion

Indian EPR is neither a clear success nor pure paper compliance. It has created a real registry, targets and a market, which did not exist before 2023. But without strong verification and a way to bring informal collection into the system, certificates risk representing paperwork more than recycled material. The critical test over the next few years is outcome data: how many tonnes physically enter formal recycling and what is recovered.

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ONE-PAGE EXAM REVISION SHEET | Case 4: India's Extended Producer Responsibility (EPR)

Must-remember facts & numbers

- EPR voluntary in 2012; mandatory in 2016; certificate-based from 1 Apr 2023.
- Dec 2024: 7,226 producers, 295 recyclers, 35 manufacturers, 53 refurbishers registered.
- Recycler capacity about 22.09 lakh t/yr vs generation 17.78 lakh t (2023-24).
- About 90% e-waste still informal; floor Rs 22/kg vs market bids as low as Rs 5.90/kg (reported).
- 2023 audit found 600,000+ fake plastic-EPR certificates (Gujarat, Maharashtra, Karnataka).
- Delhi High Court challenge to price band by major OEMs (from 2024-25).

Define EPR -> Indian evolution (2012, 2016, 2022) -> mechanism -> evidence for success -> evidence for paper compliance -> verdict -> fixes.

Memory hook: *"Register is not Recycle": counting entries is not counting tonnes.*

Key terms

Paper compliance	Meeting obligations on documents without real recycling.
Price floor	Minimum certificate price (about Rs 22/kg) set via environmental compensation band.
Informal sector	Unregistered kabadiwalas and dismantlers who handle most e-waste.
Traceability	Ability to follow material from collection to recovery.
Nil EPR target	Producer with no assigned obligation in the year.

Likely exam questions (frame your answer around these)

1. Is India's EPR a success or paper compliance? Argue with evidence.
2. What are the weaknesses of a certificate-trading EPR model?
3. Suggest measures to make EPR verifiable.

Answer skeleton

CASE STUDY 5

Bengaluru: India's E-Waste Capital?

E-Waste Case Study Report

CASESTUDYREPORT

Bengaluru: India's E-Waste Capital?

An IT hub's waste problem and the formal-informal divide

Report prepared from publicly available sources; figures are as reported by the cited sources (checked September 2026).

1. Executive Summary

Bengaluru is India's technology capital, with a very large base of IT companies, start-ups, offices and households replacing devices frequently. It is regularly described as one of India's largest e-waste generators, although by volume Mumbai and Delhi are usually ranked above it. Whether Bengaluru is "the e-waste capital" therefore depends on the definition: it is the country's most IT-driven e-waste city, but not clearly the biggest by tonnage.

The core problem is not lack of authorised capacity but weak inflow to it. The Karnataka State Pollution Control Board (KSPCB) has authorised dismantlers and recyclers, yet the formal sector reportedly gets only 10-15% of the city's e-waste; most flows through kabadiwalas and informal recyclers who use crude methods such as acid stripping and open burning. Published estimates of Bengaluru's annual e-waste vary widely, from about 20,000 tonnes to 100,000 tonnes or more, which itself shows the lack of reliable measurement.

2. Background

Bengaluru's economy is built around IT, IT-enabled services, electronics design and start-ups. The KSPCB has reported that around 1,000 tonnes of e-waste per month originates from IT companies alone. Households add to this: one recycler estimated that a middle-income Indian household generates about 25 kg of e-waste per year.

Different studies give very different city totals. An Assocham-based figure cited by Citizen Matters puts Bengaluru at roughly 20,000 tonnes a year, growing about 20% annually; campaign and NGO material has quoted 100,000 tonnes and even 2 lakh tonnes. Another widely cited claim is that Bengaluru is the thirdlargest e-waste generating city in India, after Mumbai and Delhi. Students should treat these as ranges and note the absence of a city-level measurement mechanism.

3. Strategy / Initiative

Bengaluru's response combines regulation, awareness and collection partnerships, but no single agency runs the entire chain:

- Regulatory authorisation: KSPCB authorised 27 dismantlers and recyclers under the earlier 2011 Rules to handle e-waste from the city and surroundings; since 2023 all actors must be registered on the CPCB EPR portal.

- Awareness and collection drives: KSPCB announced a month-long e-waste awareness drive across Karnataka, and NGOs such as Saahas and ENSYDE ran the bE-Responsible programme from January 2017, using Bangalore One centres as drop-off points.
- Producer Responsibility Organisations (PROs): firms such as Saahas Zero Waste help brands meet collection obligations and route waste to authorised recyclers.
- Bulk consumer duty: companies and institutions are required to hand e-waste to authorised recyclers, and the 2022 Rules extend this explicitly to bulk consumers.
- Inclusion proposals: NGOs (Hasiru Dala, Saahas) argue for identifying kabadiwalas and upgrading their skills so that they can channel material to authorised recyclers instead of banning them.

4. Impact & Results

Table 1 lists the available indicators. Because sources disagree, the ranges are shown as reported rather than reconciled.

Indicator	Reported value	Source / Year
Annual e-waste generation, Bengaluru	About 20,000 t (Assocham figure, growing about 20% a year)	Citizen Matters (Assocham)
Alternative estimate	About 1,00,000 t; other campaign material cites 2 lakh t	Citizen Matters, Saahas / ENSYDE
E-waste from IT companies	About 1,000 t per month	KSPCB, Deccan Herald
Share reaching authorised recyclers	About 10% (or 10-15%)	KSPCB officials, Deccan Herald
Authorised dismantlers / recyclers (2011 Rules)	27	KSPCB
Share of India's e-waste going to informal sector	More than 90% (widely cited)	Citizen Matters
bE-Responsible pilot collection	3 t collected; 830 kg CO2 avoided; 40,000 people reached	Bangalore One / Saahas, 2017 onward
Household generation (estimate)	About 25 kg per middle-income household per year	BinBag (recycler estimate)

Interpretation: the pilot shows that citizen collection can work, but the scale (3 tonnes) is tiny compared with the city's flows. The main finding is structural: capacity in the formal sector exists, but consumers and small businesses sell to informal dealers for quick cash, and micro and small enterprises were exempt from KSPCB authorisation requirements under earlier rules, which left a gap. Formal recyclers reported running well below licensed capacity in other cities too, suggesting the same feedstock problem.

5. SWOT Analysis

Strengths	Weaknesses
• Strong start-up and NGO ecosystem (Saahas, BinBag, Hasiru Dala) • Existing authorised recyclers with spare capacity • High digital awareness among citizens and firms	• No city-level measurement of e-waste • Only 10-15% reaches formal recyclers • Households mix small e-waste with dry waste • Exemptions for small enterprises under earlier rules
Opportunities	**Threats**
• Corporate bulk-consumer contracts with authorised recyclers • Formalising kabadiwalas as collectors • Ward-level collection points and doorstep pickups • Green jobs and urban mining	• Acid and burning-based informal recycling harms health • Rapid growth of device volumes • Consumers preferring quick cash • Illegal dumping in lakes and vacant land

6. Comparison with India

Bengaluru is one of India's most digitally advanced cities, yet it reflects the national pattern rather than an exception.

Dimension	Bengaluru	India
Formal share of ewaste	About 10-15% reaches authorised recyclers	More than 90% informal nationally (widely cited)
Data quality	Estimates range from 20,000 t to 1 lakh+ t	National generation series exists (17.78 lakh t in 2023-24), but city-level data is thin
Ranking	Often cited as third among cities	Mumbai, Delhi, Bengaluru, Chennai, Kolkata commonly ranked top
Producer role	PROs and brands run collection drives	EPR portal targets apply nationwide since 2023
Informal sector	Recognised by NGOs as a partner to be upgraded	Same debate nationally; not yet formally integrated

7. Recommendations

1. Create a city e-waste dashboard (generation, collection, recycling) and require authorised recyclers to report monthly to KSPCB.
2. Register and license kabadiwalas as aggregators with basic safety training and a fixed channel to authorised recyclers.
3. Require IT companies and other bulk consumers to publish annual e-waste disposal certificates naming their recycler.

4. Expand ward-level drop-off points and periodic doorstep pick-ups, ideally through BBMP/GBA waste systems and apartment associations.
5. Run persistent awareness campaigns emphasising the health hazards of informal acid recovery and the value of safe disposal.

8. Conclusion

Bengaluru may not be the largest e-waste generator by volume, but it is arguably the most important test case because of its IT-driven waste and its advanced NGO and recycling ecosystem. Its biggest gap is the lastmile link between waste generators and authorised recyclers. Closing that gap through better data, incentives and inclusion of informal actors would provide a model for other Indian cities.

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ONE-PAGE EXAM REVISION SHEET | Case 5: Bengaluru: India's E-Waste Capital?

Must-remember facts & numbers

- Estimates of annual city e-waste: about 20,000 t (Assocham, +20%/yr) vs about 1 lakh t (NGO material); no official city-level measure.
- KSPCB: about 1,000 t/month from IT companies; formal share only 10-15%.
- 27 authorised dismantlers/recyclers under 2011 Rules; MSEs were exempt.
- bE-Responsible (2017): Saahas + ENSYDE; Bangalore One centres; 3 t collected.
- Household estimate about 25 kg e-waste per year (middle-income).
- By volume Mumbai > Delhi > Bengaluru; Bengaluru is the IT-driven hub.

Context (IT hub) -> data ranges and caveats -> formal vs informal flows -> initiatives -> gaps -> recommendations.

Memory hook: *"Capacity is there, feedstock is not": formal plants wait while informal dealers collect.*

Key terms

KSPCB	Karnataka State Pollution Control Board: authorises recyclers.
PRO	Producer Responsibility Organisation: manages EPR on behalf of brands.
Kabadiwala	Informal scrap dealer / aggregator.
Bulk consumer	Institution or company generating large e-waste volumes.
Urban mining	Recovering metals from e-waste instead of mining ore.

Likely exam questions (frame your answer around these)

1. Is Bengaluru India's e-waste capital? Discuss with data limitations.
2. Why does the formal sector receive only 10-15% of e-waste despite available capacity?
3. Design a city-level e-waste plan for Bengaluru.

Answer skeleton

CASE STUDY 6

Mumbai's E-Waste Ecosystem

E-Waste Case Study Report

CASESTUDYREPORT

Mumbai's E-Waste Ecosystem

India's largest e-waste generating city and its informal chain

Report prepared from publicly available sources; figures are as reported by the cited sources (checked September 2026).

1. Executive Summary

Mumbai is repeatedly identified as India's largest e-waste generating city, and Maharashtra as the leading state. Its commercial density (banks, corporates, IT/ITES firms, ports) and high consumption produce large volumes, while its recycling chain is a mixture of informal collectors, dealers in trade hubs such as Dharavi, and a smaller set of authorised recyclers.

Older studies estimated Greater Mumbai's e-waste at about 11,000 to 19,000 tonnes a year in the mid-2000s, later media and industry sources cited about 96,000 tonnes, and official data for Maharashtra as a whole show 37,099 tonnes (2020-21) and 83,086 tonnes (2021-22). The figures are not comparable, but all agree on the direction: rapid growth and a chain that is mostly outside formal control. The ecosystem is best understood as a supply chain with three tiers: generators, informal collectors and dealers, and authorised recyclers.

2. Background

A 2007 Toxics Link study, "Mumbai: Choking on e-waste", found that Mumbai generated about 19,000 tonnes of e-waste a year, ahead of Delhi at 11,000 tonnes, and warned about clandestine imports from developed countries. A Maharashtra Pollution Control Board (MPCB) assessment from the same period estimated total WEEE in Maharashtra at about 20,271 tonnes, of which Greater Mumbai contributed about 11,017 tonnes, Pune about 2,584 tonnes and Navi Mumbai about 646 tonnes.

Later studies reported Mumbai as the top-ranked Indian city for e-waste, with a 2018 media analysis citing 96,000 tonnes per year and noting that banks and corporates that periodically dispose of computers and telecom equipment account for a large share. Within India, the western region accounts for about 35% of ewaste, and Maharashtra ranks first among states.

3. Strategy / Initiative

Mumbai does not have one unified program; the ecosystem is shaped by state regulation, market forces and civil society:

- Regulation by MPCB: the board licenses dismantlers and recyclers, and in 2018 the state government announced a push to enforce EPR on producers of TVs, computers and mobile phones.
- Informal collection chain: households and offices sell old devices to kabadiwalas, who pass them to dealers and dismantlers, some in Dharavi where metals are extracted and second-hand appliances refurbished.
- Authorised recycling: a few licensed units operate around Navi Mumbai and Maharashtra; one company reported a licensed capacity of 1,000 tonnes but actual processing of only 100-200 tonnes a year because of lack of feedstock (older report).

- Industry and research cooperation: from the mid-2000s, industry bodies, MPCB, NGOs and international partners (such as Empa) collaborated on assessments and pilot solutions.
- Bulk-generator focus: banks, IT/ITES firms and government offices are targeted because they generate large, concentrated volumes that can be routed to authorised recyclers under contracts.

4. Impact & Results

Table 1 collects the available numbers. The series come from different years and methods, so they should be read as indicators of scale rather than a single trend.

Indicator	Reported value	Source / Year
Mumbai WEEE generation	About 19,000 t/yr (Delhi 11,000 t/yr)	Toxics Link, 2007
Greater Mumbai share of Maharashtra WEEE	11,017 t of 20,271 t	MPCB assessment, 2007
Mumbai e-waste (later estimate)	About 96,000 t/yr	Mongabay (2018) summary
Maharashtra generation	37,099 t (2020-21); 83,086 t (2021-22)	MPCB via Lok Sabha reply
National generation, same years	13,46,496 t (2020-21); 16,01,155 t (2021-22)	CPCB via Lok Sabha reply
Maharashtra share of national (calculated)	About 2.8% (2020-21) and 5.2% (2021-22)	Author's calculation from above
Regional share of India's e-waste	West 35%, South 30%, North 21%, East 14%	Mongabay (2018)
Ranking among cities	Mumbai 1st, then Delhi, Bengaluru, Chennai, Kolkata	CWE Journal

Interpretation: MPCB's reported state figures look low compared with Mumbai-only estimates cited earlier, and this mismatch is itself evidence of unreliable estimation. A sales-and-average-life method (used by CPCB) is expected to give more consistent state-level numbers than surveys. The key point for exam answers is that Mumbai's ranking is consistent across sources, while the tonnage is not.

5. SWOT Analysis

Strengths

- Large concentrated bulk generators (banks, corporates, IT)
- Existing informal expertise in dismantling and metal recovery
- Strong recycling and refurbishing trade around Dharavi
- Active state pollution control board

Weaknesses

- Formal recyclers underused for lack of feedstock
- Poor data and inconsistent estimates
- E-waste mixed with municipal waste
- Unsafe informal processing

Opportunities	Threats
• EPR contracts with corporates • Formalising Dharavi's skills through registration and training • Deposit or buy-back schemes • Metal recovery and green jobs	• Illegal imports and clandestine trade • Air, soil and water pollution from acid recovery • Space and redevelopment pressures on informal hubs • Rapid volume growth

6. Comparison with India

Mumbai's ecosystem mirrors the national structure at a larger scale.

Dimension	Mumbai	India
Generation rank	Highest among cities; Maharashtra highest among states	India ranks among the top e-waste generators globally
Informal share	Dominant; trade hubs in Dharavi and elsewhere	More than 90% informal (widely cited)
Formal capacity	Few licensed units; underused in older reports	National capacity about 22 lakh t/yr, but collection is the bottleneck
Data	Conflicting estimates by year and method	CPCB uses sales and product-life model
Regulator posture	State government announced an EPR enforcement drive in 2018	CPCB portal and certificates since 2023

7. Recommendations

1. Create a Mumbai-wide e-waste collection network combining ward drop boxes, retailer take-back and corporate pick-ups.
2. Give informal dealers and Dharavi units a path to registration, safety equipment and access to authorised recyclers.
3. Enforce bulk-consumer rules: mandatory recycler certificates for banks, IT firms and government departments.
4. Track imports of used electronics and scrap more strictly at ports.
5. Build a single city and state e-waste dataset based on CPCB's sales and life-cycle method, with public annual updates.

8. Conclusion

Mumbai's e-waste ecosystem is both the largest and the most instructive in India: it demonstrates the scale of the challenge and the depth of informal expertise that any solution must use. Improvement will depend on closing the information gap, channelling bulk waste to authorised recyclers and integrating informal dealers with safer, formal processes.

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ONE-PAGE EXAM REVISION SHEET | Case 6: Mumbai's E-Waste Ecosystem

Must-remember facts & numbers

- Mumbai is #1 city and Maharashtra #1 state for e-waste generation.
- Toxics Link 2007: Mumbai 19,000 t vs Delhi 11,000 t; MPCB 2007: Greater Mumbai 11,017 t of Maharashtra's 20,271 t.
- Maharashtra (MPCB): 37,099 t (2020-21), 83,086 t (2021-22); India 13.46 and 16.01 lakh t.
- Later estimate for Mumbai: about 96,000 t/yr (2018 media).
- Regional share: West 35%, South 30%, North 21%, East 14%.
- Authorised recycler example: licensed 1,000 t but processed 100-200 t/yr (feedstock shortage).

Scale and ranking -> three-tier chain (generators / informal / formal) -> data with caveats -> policy actions -> recommendations.

Memory hook: *"Big city, small pipe": huge volumes, narrow formal channel.*

Key terms

WEEE	Waste Electrical and Electronic Equipment.
MPCB	Maharashtra Pollution Control Board.
Clandestine imports	Illegal shipments of used electronics from other countries.
Bulk generators	Banks, corporates and offices generating large e-waste volumes.
Feedstock	E-waste supply reaching a recycler.

Likely exam questions (frame your answer around these)

1. Describe the structure of Mumbai's e-waste ecosystem.
2. Why do e-waste estimates for Mumbai differ so much?
3. How can Mumbai bring bulk generators and informal dealers into a formal system?

Answer skeleton

CASE STUDY 7

Kerala's Clean Kerala Company

E-Waste Case Study Report

CASE STUDY REPORT

Kerala's Clean Kerala Company

A state-owned company and women-led collectors for non-biodegradable waste

Report prepared from publicly available sources; figures are as reported by the cited sources (checked September 2026).

1. Executive Summary

Clean Kerala Company Limited (CKCL) is a state-owned company under the Local Self-Government Department. It works with local bodies and the Haritha Karma Sena (HKS), a Kudumbashree-supported workforce of women, to collect, aggregate and dispose of non-biodegradable waste, including plastics, textiles, glass and e-waste. The model puts a public company at the aggregation and processing end and community workers at the doorstep.

Growth has been rapid. Reports show waste received by CKCL rising from 7,657 tonnes in 2021-22 to 30,217 tonnes in 2022-23, and HKS collecting 50,190 tonnes in 2024-25 and 86,295.7 tonnes in 2025-26, with 37,481 women in 4,438 units earning Rs 11.49 crore from supplying waste in that year. Gaps remain: limited in-state recycling capacity means much recyclable waste goes to Tamil Nadu, and door-to-door coverage is uneven across local bodies.

2. Background

Kerala has a high population density, high literacy and heavy consumption of packaged goods, but limited land for landfills and strong public opposition to dumping sites. The state built a decentralised waste system in which local governments (panchayats and municipalities) are responsible for collection, the Suchitwa Mission gives technical support, the Haritha Keralam Mission handles community engagement, Kudumbashree provides livelihood support to Haritha Karma Sena workers, and CKCL takes over nonbiodegradable and legacy waste.

Floods in 2018 and the Brahmapuram fire in 2023 highlighted the need for strong non-biodegradable waste management. After the 2018 floods, CKCL was assigned the collection of e-waste and coordinated with a Hyderabad-based recycler. The company also collected about 750 tonnes from Kochi in two months following the Brahmapuram incident.

3. Strategy / Initiative

The Kerala model has five working parts:

- Doorstep collection by Haritha Karma Sena: authorised women collectors visit households and shops for non-biodegradable waste and charge a user fee; HKS members were over 31,000 in 2023 and 37,481 in 2025-26.
- Material collection facilities: local bodies run 1,000+ Material Collection Facilities (MCFs) and over 100 Resource Recovery Facilities (RRFs) with shredding and baling, as reported to the NGT.

- CKCL as aggregator and disposer: it buys segregated waste from HKS, shreds and sells plastics for road tarring and other uses, and arranges recycling of items like textiles, glass and e-waste.
- Special streams: separate handling of items such as tablet (blister) strips, which are sent to a biomedical waste plant in Kochi (7.74 tonnes in 2022-23).
- Livelihood and incentive design: compensation and user fees for HKS, growth of over 400 microenterprises, and a proposed incentive for households in Kollam that segregate and hand over dry waste.

4. Impact & Results

Table 1 presents the main quantitative results from government submissions and media reports.

Indicator	Result	Year / Source
Waste received by CKCL	7,657 t (2021-22); 30,217 t (2022-23)	Onmanorama, 2023
HKS non-biodegradable collection	50,190 t (2024-25); 86,295.7 t (2025-26)	UNI, Aug 2026
HKS workforce	31,000 members (2023); 37,481 women in 4,438 units (2025-26)	Onmanorama; UNI
HKS income from supplying waste	Rs 11.49 crore	2025-26
Compensation paid to HKS	Rs 2.60 crore (2021-22) to Rs 5.08 crore (2022-23)	Onmanorama, 2023
Micro-enterprises linked to HKS	More than 400	UNI, 2026
Shredded plastic produced / sold	3,293 t / 2,987 t	Kerala reply to NGT
Polymerised road built with plastic	5,285.54 km	Kerala reply to NGT
Kollam district revenue from recycling	Over Rs 29 crore from 2,797 t of inorganic waste	2025-26, Onmanorama
Facilities	1,201 MCFs, 149 RRFs, only 123 registered small plastic recycling units in the state	Kerala reply to NGT

Interpretation: the trend line is strongly positive on collection and livelihoods; the income per worker remains modest (Rs 11.49 crore across more than 37,000 women, or roughly Rs 3,000 per worker per year from supply of waste alone, before user fees; author's calculation). The state's weakest link is recycling capacity, since most recyclables travel out of state and non-recyclable plastics are used for roads or co-processing.

5. SWOT Analysis

Strengths	Weaknesses
• Decentralised model with strong local government and Kudumbashree links • Women's livelihoods and micro-enterprises • Rapid growth in tonnage • Road-tarring outlet for non-recyclable plastics	• Limited in-state recycling units • Uneven door-to-door coverage (some ULBs below 25%) • Dependence on user fees and CKCL purchase prices • E-waste is a small part of the collected mix
Opportunities	**Threats**
• EPR tie-ups with producers • Household incentives for segregation • More recycling units and industrial space • Formal e-waste and textile streams	• Fluctuating scrap prices • Cement-plant co-processing not fully available • Fires and landfill crises • Financial sustainability of HKS units

6. Comparison with India

Kerala is a strong model for collection and social inclusion, but it is more focused on general nonbiodegradable waste than on e-waste alone.

Dimension	Kerala	India
Institutional model	State-owned CKCL plus local bodies plus HKS	Most states rely on municipal contracts or private operators
Workforce	Formalised women's collectives (Kudumbashree)	Waste pickers usually informal; Pune SWaCH is a comparable exception
Financing	User fees, CKCL payments, scrap revenue	Varies; user-fee recovery is weak in many cities
E-waste	Collected as part of non-biodegradable stream; sent to registered recyclers outside the state	National EPR system through registered recyclers
Weakness	In-state recycling capacity	Collection and formalisation gaps

7. Recommendations

1. Add dedicated e-waste and battery collection days through HKS, paid for through producer EPR contracts.
2. Attract recycling investment into Kerala's industrial areas, so that recyclables are processed in-state.
3. Publish ward-level coverage and segregation data to raise weak-performing local bodies to full doorto-door coverage.
4. Extend household incentives (as proposed in Kollam) to other districts, tied to verified segregation.
5. Protect HKS incomes through minimum floor prices for common materials and health cover.

8. Conclusion

Clean Kerala Company shows that a public company can link community workers with recyclers and roadbuilding outlets to move large volumes of non-biodegradable waste. Its lesson for India is the combination of institutional clarity and livelihood design. To become a full circular model, Kerala must now build recycling capacity within the state and deepen its work on e-waste through producer partnerships.

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ONE-PAGE EXAM REVISION SHEET | Case 7: Kerala's Clean Kerala Company

Must-remember facts & numbers

- CKCL: state company under LSGD; partners Suchitwa Mission, Haritha Keralam Mission, Kudumbashree.
- HKS collection: 50,190 t (2024-25); 86,295.7 t (2025-26); 37,481 women in 4,438 units; earned Rs 11.49 crore.
- CKCL received 7,657 t (2021-22) and 30,217 t (2022-23).
- 1,201 MCFs, 149 RRFs but only 123 small registered plastic recyclers in state; most recyclables go to Tamil Nadu.
- 5,285.54 km of plastic-polymerised roads; shredded plastic 3,293 t produced.
- Kollam 2025-26: 2,797 t inorganic waste, over Rs 29 crore.

Institutions -> doorstep collection -> aggregation and disposal (CKCL) -> tonnage growth -> livelihoods -> gaps (recycling capacity) -> recommendations.

Memory hook: *"Sena collects, Company aggregates, Road absorbs, Recycler missing."*

Key terms

Haritha Karma Sena (HKS)	Kudumbashree-backed women's workforce doing doorstep collection.
MCF / RRF	Material Collection Facility / Resource Recovery Facility.
Kudumbashree	Kerala state poverty-eradication and women's empowerment mission.
Polymerised road	Road built using shredded plastic mixed with bitumen.
ULB	Urban Local Body.

Likely exam questions (frame your answer around these)

1. Explain the roles of CKCL, HKS and local governments in Kerala's waste system.
2. Assess strengths and limitations of the Kerala model.
3. How can Kerala improve e-waste handling?

Answer skeleton

CASE STUDY 8

Pune's SWaCH Model

E-Waste Case Study Report

CASE STUDY REPORT

Pune's SWaCH Model

India's first wholly owned cooperative of waste pickers

Report prepared from publicly available sources; figures are as reported by the cited sources (checked September 2026).

1. Executive Summary

SWaCH (Solid Waste Collection and Handling) is a member-owned cooperative of self-employed waste pickers and other urban poor workers, authorised by the Pune Municipal Corporation (PMC) to provide doorto-door collection. It grew out of the waste pickers' trade union KKPKP (formed in 1993), signed its first agreement with PMC in 2008, and is often cited as a global example of integrating informal workers into municipal services.

SWaCH's own 2025 figures state that it covers about 10 lakh households daily, integrates more than 4,000 waste pickers, recycles 82,891 tonnes of waste annually and has saved PMC Rs 113 crore. The model is financed mainly by household user fees and scrap sales, with PMC providing infrastructure and support. It shows that formalising waste pickers can improve service, livelihoods and recovery rates, though it also needs strong contracts, timely municipal payments and citizens who segregate.

2. Background

In the early 1990s, waste pickers in Pune, mostly women from marginalised communities, worked in dumps and streets, facing stigma, low income and hazards. Adult-education workers at SNDT organised them, leading to the union Kagad Kach Patra Kashtakari Panchayat (KKPKP) in 1993, with the goal of obtaining recognition as workers, identity cards and better earnings.

Pune, like many Indian cities, faced open dumping, unsafe conditions and heavy landfilling. In 2005-2007 the union created SWaCH as a workers' cooperative and in 2008 signed an agreement with PMC. A pilot covered about 1,500 workers. In 2013 workers operated without municipal support while negotiating over delayed benefits and equipment, and a second agreement was signed in 2016. Over 1.2 million people (about a third of Pune's population) lived in slums with little or no waste services, which gave the model a clear publicservice purpose.

3. Strategy / Initiative

The SWaCH model can be described through its operating design:

- Cooperative ownership: waste pickers are members and participate in governance, rather than being contractors or daily-wage employees.
- Door-to-door collection: pairs of members serve about 150-400 households each, collecting segregated waste and transferring it to city collection vehicles.

- User fees and scrap income: households and commercial clients pay directly, and members earn from selling recyclables; PMC supports administrative staff, equipment and health benefits.
- Segregation at source: SWaCH members encourage households to separate wet and dry waste and help maintain compost pits in large residential societies.
- Service diversification: composting and biogas services for bulk generators, the V-Collect programme for old electronics and household items, Red Dot sanitary-waste awareness, and event waste collection.
- Partnership and negotiation: written agreements with PMC (2008, 2016) define obligations, and the cooperative negotiates for timely payments.

4. Impact & Results

Table 1 combines SWaCH's own 2025 impact statement with earlier independent and press reports. Figures come from different years, so they show scale rather than a single consistent series.

Indicator	Result	Year / Source
Pilot workers under PMC agreement	About 1,500	2008, Prize for Cities
Membership	3,500+ (2018); 3,900+ (later); 4,000+ (2025)	Prize for Cities; CENFA; SWaCH
Households served	About 4 lakh (mid-2010s); 6 lakh; 10 lakh (2025 claim)	Down To Earth; UNEP; SWaCH
Waste recycled	More than 50,000 t/yr (UNEP); 82,891 t annually (2025)	UNEP; SWaCH
Daily collection (earlier)	About 600 t/day, of which 140 t dry	Down To Earth
Recyclables diverted daily	About 90 t/day	CENFA
Saving to PMC	Rs 113 crore (SWaCH 2025); about US$316,455/yr in transport	SWaCH; CENFA
Environmental benefit (paper)	Over 350,000 trees saved; 130,000 t CO2e avoided per year	UNEP
Waste arriving segregated	About 43%	PMC official, Down To Earth

Interpretation: the strongest evidence is for livelihoods and service coverage; the environmental figures come from SWaCH and UNEP summaries and should be cited as reported, not audited. Household coverage numbers rise over time and differ by source, so it is safest to quote the source and year. The 43% segregation figure shows that even the best-known model depends on citizen behaviour.

5. SWOT Analysis

Strengths	Weaknesses
• Member-owned, worker-governed model • Formal agreement with the municipality • Trusted doorstep relationship with households • Recognised globally as a pro-poor model	• Dependence on PMC payments and contract renewal • Only about 43% of waste arrives segregated • Member earnings vary with scrap prices • Limited processing infrastructure of its own
Opportunities	**Threats**
• Scaling to e-waste and other EPR streams • Composting and biogas services for bulk generators • Replication in other cities • Carbon and plastic-credit revenue	• Privatisation or centralised contracts that exclude pickers • Delayed municipal payments • Competition from private operators • Falling scrap prices

6. Comparison with India

Pune SWaCH is itself an Indian model; the comparison is with typical Indian practice and with Kerala's Haritha Karma Sena.

Dimension	Pune SWaCH	India
Worker status	Cooperative members with ID, governance role and benefits	Most waste pickers are informal and unrecognised
Ownership	Wholly owned by waste pickers	Kerala HKS is state-supported and Kudumbashree-led; many cities use private contractors
Financing	User fees plus scrap plus PMC support	Often municipal budgets with weak fee collection
Scale	About 10 lakh households (2025 claim), 70% of Pune per SWaCH	Coverage varies; door-to-door under 25% in some Kerala local bodies
Weak point	Segregation and municipal payment delays	Formalisation and recognition of waste pickers

7. Recommendations

1. Incorporate SWaCH-like cooperatives in EPR systems, so that producers pay them to collect dry waste and e-waste.
2. Guarantee timely payment through escrowed municipal funds and standard contracts with clear service levels.
3. Support segregation through household incentives and enforcement, with the goal of raising the segregated share well above 43%.
4. Provide cooperatives with sorting sheds, balers and compost or biogas units so that more value is captured locally.

5. Adopt a national policy that recognises waste pickers with ID cards, social security and priority for municipal contracts.

8. Conclusion

SWaCH shows that waste pickers can be partners rather than a problem for city waste systems. Its lessons are member ownership, contractual recognition and fee-based finance. Its continued success depends on fair payment terms and citizen participation, and its next opportunity is to become the collection arm for e-waste and other EPR streams.

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ONE-PAGE EXAM REVISION SHEET | Case 8: Pune's SWaCH Model

Must-remember facts & numbers

- KKPKP (union) formed 1993; SWaCH cooperative set up mid-2000s; first PMC agreement 2008 (about 1,500 workers); second agreement 2016.
- 2025 claim: 10 lakh households, 4,000+ pickers, 82,891 t recycled/yr, Rs 113 crore saved to PMC.
- Earlier: about 600 t/day collected (140 t dry); about 43% segregated.
- Pairs of workers serve 150-400 households each; earnings from user fees plus scrap.
- UNEP: 350,000 trees and 130,000 t CO2e avoided per year from paper recycling.
- Extra services: composting/biogas, V-Collect (e-items), Red Dot (sanitary waste).

Union origin -> cooperative model -> PMC agreements -> operations and finance -> impact data -> risks -> replication and EPR link.

Memory hook: *"Own it, Contract it, Fee it, Scale it."*

Key terms

SWaCH	Solid Waste Collection and Handling cooperative.
KKPKP	Kagad Kach Patra Kashtakari Panchayat: waste pickers' trade union.
PMC	Pune Municipal Corporation.
User fee model	Households pay collectors directly for doorstep service.
YIMBY	"Yes In My Back Yard": decentralised waste processing.

Likely exam questions (frame your answer around these)

1. Explain the SWaCH model and how it benefits waste pickers and the city.
2. What conditions are needed to replicate SWaCH elsewhere?
3. How could SWaCH be integrated with e-waste EPR?

Answer skeleton

CASE STUDY 9

Dharavi Recycling Ecosystem

E-Waste Case Study Report

CASESTUDYREPORT **Dharavi**

Recycling Ecosystem

The informal engine of Mumbai's recycling and its uncertain future

Report prepared from publicly available sources; figures are as reported by the cited sources (checked September 2026).

1. Executive Summary

Dharavi, in central Mumbai, houses one of the world's densest informal recycling economies. Thousands of small units, many concentrated in the area known as the 13th Compound, sort, grind, wash and reprocess plastics, and also handle metals, paper, cloth, glass and second-hand goods and e-waste. Waste collected across the city is sold up a chain of pickers, kabadiwalas and processors, and comes back to industry as flakes, granules and secondhand goods.

The scale is large but poorly measured: estimates cited include more than 5,000 single-room recycling units, up to 60-80% of Mumbai's plastic passing through Dharavi (reported estimates), and about 6,000 tonnes of plastic recycled per month in the 13 compounds (one trader's estimate). The system offers livelihoods and low-cost material recovery but operates without formal safety, pay or environmental controls, and its future is uncertain because of redevelopment plans for the land.

2. Background

Dharavi began in 1882 when polluting industries were moved to marshy land at the edge of colonial Bombay. Over time, it attracted successive waves of migrants. Sources describe an area of roughly 520 to 600 acres (2.1 sq km) with around a million residents, sitting on very valuable land between Mumbai's business districts. The 13th Compound was reclaimed from the Mithi River estuary and grew with textile-mill scrap and, later, plastics.

The value chain is layered: waste pickers walk long distances to collect 40 to 50 kg of waste a day, sell to kabadiwalas, and material is consolidated and trucked to Dharavi. A sack of unsorted plastic may sell at Rs 1015 per kg, but value rises sharply once it is sorted by type and colour and ground into flakes. Workers doing sorting may earn about Rs 300-500 per day, and units often make or repair their own machines locally. **3.**

Strategy / Initiative

Dharavi's ecosystem is not a planned strategy; it is a self-organised market. Its main operating features are:

- Multi-tier supply chain: waste pickers, then kabadiwalas, then sorting units, then crushers and washing units, then pellet or flake buyers such as packaging and manufacturing firms.
- Extreme specialisation: units focus on a material (plastic by grade and colour, cardboard, metal, cloth, glass, wood), passing rejects to other units.
- Labour-intensive sorting: workers grade plastics by sound and touch into several grades, which raises the value of recovered material.
- Local machinery and skills: crushers, grinders and blow-moulders are often built or repaired inside Dharavi.

- Second-hand and e-waste trade: recovery of metals from e-waste and refurbishing of second-hand appliances happen alongside plastics.
- Community networks: workers often migrate from the same villages, so labour and credit flow through personal links rather than formal contracts.

4. Impact & Results

The figures in Table 1 are estimates from journalism, NGOs and academic sources; there is no audited count of Dharavi's recycling output, and the numbers should be quoted as estimates.

Indicator	Reported estimate	Source
Recycling units	More than 5,000 single-room units	ACORN India
Area	About 520 acres (ACORN) to 600 acres (2026 feature); 2.1 sq km (academic)	Various
Share of Mumbai's plastic processed	Up to 60-80% (estimates)	2026 feature; other sources
Plastic recycled in 13th Compound	About 6,000 t per month (one trader's estimate)	Revolve / Circular, 2024
Typical unit throughput	20+ t of plastic per month (one unit manager)	Revolve / Circular, 2024
Waste picker collection	40-50 kg per day, walking 15-20 km	Revolve / Circular, 2024
Price uplift from sorting	Unsorted plastic Rs 10-15 per kg; sorted material sells for more	Revolve / Circular, 2024
Worker earnings	Rs 300-500 per day (sorting); Rs 10,000 per month (older report)	2026 feature; ACORN
Claim: share of Mumbai's solid waste recycled	About 80% (unverified media claim)	City Monitor

Interpretation: the numbers show a large informal recovery system that saves the city transport and landfill costs and creates jobs. However, the highest claims (such as 80% of Mumbai's waste) come from journalistic sources and are not verified, so an exam answer should say "reported estimates". Academic work describes waste workers as "human infrastructure" that reduces the plastic entering landfill and marine environments, while also noting hazards, low pay and lack of recognition.

5. SWOT Analysis

Strengths	Weaknesses
• Very high material recovery at very low cost • Deep skills in grading and reprocessing • Jobs and entry-level livelihoods for migrants • Local machine building and repair	• No formal safety, pay or environmental controls • Cramped conditions and health hazards • No verified data or certification • Dependence on middlemen for prices
Opportunities	**Threats**
• Registration and EPR partnerships with brands • Training, safety equipment and shared washing/effluent facilities • Certified recycled-plastic supply chains • Green-collar worker recognition	• Land redevelopment displacing units • Regulatory crackdowns without alternatives • Falling scrap prices and cheap virgin plastic • Pollution from unregulated washing and burning

6. Comparison with India

Dharavi represents in concentrated form the informal recycling economy that handles most of India's dry waste and e-waste.

Dimension	Dharavi	India
Structure	Self-organised, multi-tier private market	Nationally, informal collection and dismantling dominate
Recognition	Unregistered units; no formal status	EPR and e-waste rules recognise only registered recyclers
Safety	Little protective equipment; hazardous exposures	Rules require environmentally sound processes at registered plants
Data	Estimates from journalists and NGOs	CPCB portal covers registered entities only
Contrast with cooperatives	Workers earn wages or piece rates from owners	SWaCH gives pickers ownership; Kerala HKS gives state support

7. Recommendations

1. Recognise Dharavi units through a phased registration system with simple compliance steps and hand-holding rather than closure.
2. Link brands and producers to Dharavi through verified buy-back contracts that generate EPR credits for recycled plastic.
3. Provide shared infrastructure: common effluent treatment, safe washing lines, fire safety and worker protective equipment.
4. Commission an independent survey to measure actual tonnage, jobs and income, so that policy uses facts instead of estimates.
5. In any redevelopment, reserve space and continuity for recycling functions instead of assuming they can be relocated at will.

8. Conclusion

Dharavi is the clearest example of how much material recovery India already achieves without formal recycling policy, and of the price paid by workers. A responsible solution would neither romanticise nor eliminate it: it should measure it, improve safety and integrate it with EPR and formal supply chains, so that the skills and livelihoods survive redevelopment.

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ONE-PAGE EXAM REVISION SHEET | Case 9: Dharavi Recycling Ecosystem

Must-remember facts & numbers

- Dharavi founded 1882; area about 520-600 acres (2.1 sq km); about 1 million people.
- More than 5,000 single-room recycling units; core: 13th Compound.
- Estimated 60-80% of Mumbai's plastic passes through Dharavi (estimates, not audited).
- Trader estimate: about 6,000 t plastic/month in 13 compounds.
- Unsorted plastic Rs 10-15/kg; sorting into about seven grades raises value.
- Worker pay about Rs 300-500/day; redevelopment threatens the ecosystem.

History and scale -> value chain -> data (as estimates) -> benefits and hazards -> redevelopment risk -> formalisation options.

Memory hook: *"Pick, Sort, Grind, Wash, Sell": five steps of the Dharavi chain.*

Key terms

13th Compound	Cluster of plastic recycling units in Dharavi.
Kabadiwala	Scrap dealer who buys from pickers and sells onward.
Informal sector	Unregistered economic activity outside formal regulation.
Human infrastructure	Workers whose labour functions as waste system infrastructure.
Green-collar worker	Term for recognising waste workers' environmental role.

Likely exam questions (frame your answer around these)

1. Describe the Dharavi recycling value chain from pickers to reprocessors.
2. Critically evaluate: should Dharavi's recycling be formalised or replaced?
3. How can Dharavi be linked with EPR?

Answer skeleton

CASE STUDY 10

Dell's Circular Economy Strategy

E-Waste Case Study Report

CASESTUDYREPORT Dell's

Circular Economy Strategy

How a global electronics producer designs for take-back, reuse and recycled content

Report prepared from publicly available sources; figures are as reported by the cited sources (checked September 2026).

1. Executive Summary

Dell Technologies has organised its sustainability work around a circular economy commitment for 2030: reuse or recycle one metric ton of product for every metric ton customers buy, make more than half of product content from recycled, renewable or reduced-carbon materials, and make 100% of packaging from recycled or renewable material or reused packaging. The programme combines design, materials sourcing, take-back and refurbishment.

Dell reports steady progress, especially in packaging (packaging figures above 90% have been reported) and take-back (more than 2 billion pounds of used electronics recovered since 2007). Product-content targets are harder because electronics use complex materials that are difficult to recycle without losing quality. For India, the case illustrates how a producer can go beyond legal EPR compliance by using design and take-back as competitive strategy.

2. Background

Global e-waste is growing: about 53.6 million tonnes in 2019, projected to reach 74 million tonnes by 2030, with only 17.4% formally collected and recycled in 2019. As producers face EPR laws worldwide and customers demand lower-carbon IT, circular strategies are becoming a business issue as well as an environmental one.

Dell started asset recovery services for commercial customers in 1996 and began using recycled materials in products in the late 1990s. It reports achieving its 2020 goal of using 100 million pounds of recycled content ahead of schedule. In 2020 it announced its 2030 "moonshot" goals, which extend from product design to end-of-life. Dell is also a member of the Circular Electronics Partnership.

3. Strategy / Initiative

Dell's strategy has four connected parts:

- Design for circularity: simpler products with fewer materials, modular and accessible components for repair and easier disassembly, and prototypes such as Concept Luna that explore easily replaceable components.
- Recycled and renewable materials: recycled aluminium, steel, magnesium and copper; recycled cobalt (first use of 50% recycled cobalt in commercial notebooks in 2024); reclaimed carbon fibre; bio-based plastics; and ocean-bound plastic, with about 71% recycled or renewable content reported for the lids of certain Latitude and Precision models.
- Sustainable packaging: bamboo, recycled paper pulp, sugarcane fibre and recycled plastic, with packaging for commercial laptops now reported as 100% recycled or renewable material.

- Take-back and reuse: Asset Recovery Services for any brand of equipment, partnerships with more than 2,000 Goodwill locations in the US, and Dell Financial Services leasing, through which about 95% of returned devices are refurbished (as reported in India).
- Repair and lifetime extension: spare parts and repair tutorials to keep systems in use longer.

4. Impact & Results

Table 1 lists reported progress. Most figures are company-reported and use Dell's own definitions, so they should be quoted as "Dell reports".

Indicator	Dell-reported result	Year / Source
Used electronics recovered since 2007	More than 2 billion pounds	Business Today interview, 2021
Material recovered via Goodwill partnership	About 620 million pounds	Waste360, FY2023
Sustainable materials used in products	More than 86 million pounds	FY2023
2020 recycled-content goal	100 million pounds, achieved early	Dell blog
Packaging recycled or renewable	85% (India interview, 2021); 96.4% (later summary)	Business Today; Tech for Good
Devices refurbished from DFS returns	About 95%	Dell India, 2021
Ocean-bound plastic	More than 443,000 pounds saved (Dell website)	Dell sustainability page
Product content (2030 goal)	Target above 50%; consult latest Dell ESG report for current percentage	Dell 2030 goals

Interpretation: packaging is easy to improve compared with product content, because electronics need tightly specified metals and plastics. Dell's metric of "one ton reused or recycled for every ton sold" is also defined by weight and does not measure how much is actually recovered from customers' old equipment. Company sustainability data should therefore be cross-checked against third-party assurance or the latest ESG report.

5. SWOT Analysis

Strengths

- Long record: asset recovery since 1996
- Global take-back and refurbishing network
- Measurable 2030 goals
- Design work on modularity and repair

Weaknesses

- Company-reported data with own definitions
- Product recycled content still far below the 50% goal (as reported in earlier years)
- Recycling collections depend on customer return behaviour

Opportunities	Threats
• Growth in refurbished and leasing markets • India's EPR and bulk-consumer rules create demand for compliant take-back • Closed-loop recovery of metals and plastics	• Complex materials reduce recycling quality • Scarce supply of recycled inputs • Competitors and consumers' short upgrade cycles • Greenwashing risk if claims are not verified

6. Comparison with India

The comparison is between Dell's voluntary, producer-led approach and the mostly compliance-driven, informal e-waste landscape in India.

Dimension	Dell	India
Driver	Corporate goals plus customer demand plus EPR laws	Regulation (EPR) is the main driver; voluntary initiatives are limited
Design	Modularity, repair and recycled content	Few Indian OEMs report circular design metrics
Take-back	Asset recovery for commercial customers; refurbishing first	Mostly informal resale and dismantling; bulk consumers now required to use authorised recyclers
Refurbishing	About 95% of leased devices returned to DFS reported as refurbished	Refurbishing certificates recognised under 2022 Rules but small scale
Transparency	Annual ESG reporting (self-reported)	CPCB portal data; limited outcome disclosure

7. Recommendations

1. Indian producers should publish annual circularity reports covering take-back tonnes, refurbishing share and recycled content by product line.
2. Use bulk-consumer channels (corporates, government IT) for structured take-back, following Dell's commercial asset recovery model.
3. Make refurbishing certificates more attractive through simple verification and links to leasing schemes.
4. Support repair by providing spare parts and manuals to independent repairers.
5. Require independent assurance of circular economy claims to prevent greenwashing.

8. Conclusion

Dell's strategy shows how a producer can integrate design, materials, packaging and take-back into one circular plan. Its strengths are the long track record, the refurbish-first approach and clear 2030 goals; its limits are self-reported data and the difficulty of raising recycled content in complex electronics. India's policy should encourage such corporate strategies through verified reporting, while its EPR system ensures that all producers, not only leaders, meet minimum standards.

9. References

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ONE-PAGE EXAM REVISION SHEET | Case 10: Dell's Circular Economy Strategy

Must-remember facts & numbers

- 2030 goals: 1 t reused/recycled per 1 t sold; 50%+ product content recycled/renewable/low-carbon; 100% packaging recycled/renewable/reused.
- Asset recovery services since 1996; 2 billion+ lb recovered since 2007.
- FY2023: 86 million lb sustainable materials; about 620 million lb recovered via Goodwill partnership.
- First use of 50% recycled cobalt in commercial notebooks (2024).
- About 95% of DFS-returned devices refurbished (India interview, 2021).
- Global e-waste: 53.6 Mt (2019) to 74 Mt (2030); 17.4% formally recycled (2019).

Global e-waste context -> Dell's three goals -> four strategy pillars (design, materials, packaging, take-back) -> results (reported) -> limits -> India lessons.

Memory hook: *"Design it, Source it, Pack it, Take it back."*

Key terms

Circular economy	System that keeps materials in use through reuse, repair, refurbishing and recycling.
DFS	Dell Financial Services: leasing that brings devices back in good condition.
PCR plastic	Post-consumer recycled plastic.
Closed-loop	Material from old products returns to make new ones.
Moonshot goal	Dell's term for its ambitious 2030 targets.

Likely exam questions (frame your answer around these)

1. Explain Dell's 2030 circular economy goals and strategies.
2. Discuss limits of company-reported circularity data.
3. How can Indian producers adopt a Dell-style circular strategy?

Answer skeleton